



Capture The Moment

In an age where digital memories rule the roost, the camcorder has become an indispensable tool. We demystify the tech behind the magic

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The time is prehistoric. An unkempt, unshaven caveman reaches for the nearest flintstone and with the skill of a present-day four-year-old draws the previous day's successful hunt on the walls.

Echoing a distant future wherein memories, cherished or otherwise, are stored and exchanged over bits of paper and plastic. As was then and is today, the message and the sentiment remains the same—capturing a memory for future retrieval. The change is in the medium. Presenting a closer look at one such medium—the camcorder.

A camcorder consists of three main elements. To begin with, you have the lens mechanism, used to gather light and focus it internally. The mechanism consists of a series of lenses, and motors to provide focus and zoom. Finally, an iris regulates the amount of light gathered. Next in line is the Charge Coupled Device or CCD, an electronic element about one cm in size consisting of anywhere between three lakh to five lakh light-sensitive cells. These are basically photo-sensitive diodes.

Finally, there is the recording section of a camcorder, which records everything seen through the lens, on a variety of media. Apart from these three areas, there is also the viewfinder, which is usually an LCD panel in modern camcorders. The viewfinder shows you

everything the lens sees, helping you frame a shot to record what is needed.

How do these disparate elements record your child's first birthday party or your company's show-stopping presentation?

At the heart of the system beats the CCD. The lens mechanism gathers light and focuses an inverted image onto the CCD element. Light is made up of particles of energy termed photons—the energy of the photon is directly in proportion to the intensity of light. A brightly-lit surface will send forth photons full of energy, while photons from a dark area are low on energy. When

both these types of photon reach the photo-sensitive cells of the CCD, their light energy is transferred to the cells in the form of electrical charges: a stronger charge from an energetic photon, a weaker charge from a tired one.

An arrangement such as this will form a black and white image with contrasting dark and light areas, much like an artist's charcoal-drawn impressions of a scene. The colour information of the scene shot by the lens needs to be translated as well.

Since any shade of colour can be reproduced by mixing the three primary colours—red, blue and green—a camcorder only needs to record the varying levels of these primary colours in a scene to faithfully recreate the colours.

This is done in either of two ways; the path taken by a camcorder determines its cost. The costlier method is to use three separate CCDs, one for each of the primary colours. Thus, a red CCD records the variation of the colour red in a scene—similarly for blue and green. The three separate colour variation maps are then internally combined (as an overlay) to form a colour image. By using three CCDs, the cost increases. Such camcorders are called three-chip camcorders.

The cheaper alternative is to divide a solitary CCD into permanent colour filters of red, green and blue. These can be arranged in a variety of manner on a CCD: the information captured in



MiniDV camcorders are the most widely available

